



Level 3 Certificate MATHEMATICAL STUDIES 1350/1

Paper 1

Mark scheme

June 2025

Version: 1.0 Final



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

No student should be disadvantaged on the basis of their gender identity and/or how they refer to the gender identity of others in their exam responses.

A consistent use of 'they/them' as a singular and pronouns beyond 'she/her' or 'he/him' will be credited in exam responses in line with existing mark scheme criteria.

Further copies of this mark scheme are available from aqa.org.uk

Copyright information

AQA retains the copyright on all its publications. However, registered schools/colleges for AQA are permitted to copy material from this booklet for their own internal use, with the following important exception: AQA cannot give permission to schools/colleges to photocopy any material that is acknowledged to a third party even for internal use within the centre.

Copyright © 2025 AQA and its licensors. All rights reserved.

Glossary for Mark Schemes

Mathematical Studies examinations are marked in such a way as to award positive achievement wherever possible. Thus, for Mathematical Studies papers, marks are awarded under various categories.

If a student uses a method which is not explicitly covered by the mark scheme the same principles of marking should be applied. Credit should be given to any valid methods. Examiners should seek advice from their senior examiner if in any doubt.

M	Method marks are awarded for a correct method which could lead to a correct answer.
A	Accuracy marks are awarded when following on from a correct method. It is not necessary to always see the method. This can be implied.
B	Marks awarded independent of method.
ft	Follow through marks. Marks awarded for correct working following a mistake in an earlier step.
SC	Special case. Marks awarded for a common misinterpretation which has some mathematical worth.
M dep	A method mark dependent on a previous method mark being awarded.
B dep	A mark that can only be awarded if a previous independent mark has been awarded.
oe	Or equivalent. Accept answers that are equivalent. eg accept 0.5 as well as $\frac{1}{2}$
[a, b]	Accept values between a and b inclusive.
[a, b)	Accept values $a \leq \text{value} < b$
3.14 ...	Accept answers which begin 3.14 eg 3.14, 3.142, 3.1416
Use of brackets	It is not necessary to see the bracketed work to award the marks.

Examiners should consistently apply the following principles

Diagrams

Diagrams that have working on them should be treated like normal responses. If a diagram has been written on but the correct response is within the answer space, the work within the answer space should be marked. Working on diagrams that contradicts work within the answer space is not to be considered as choice but as working, and is not, therefore, penalised.

Responses which appear to come from incorrect methods

Whenever there is doubt as to whether a student has used an incorrect method to obtain an answer, as a general principle, the benefit of doubt must be given to the student. In cases where there is no doubt that the answer has come from incorrect working then the student should be penalised.

Questions which ask students to show working

Instructions on marking will be given but usually marks are not awarded to students who show no working.

Questions which do not ask students to show working

As a general principle, a correct response is awarded full marks.

Misread or miscopy

Students often copy values from a question incorrectly. If the examiner thinks that the student has made a genuine misread, then only the accuracy marks (A or B marks), up to a maximum of 2 marks are penalised. The method marks can still be awarded.

Further work

Once the correct answer has been seen, further working may be ignored unless it goes on to contradict the correct answer.

Choice

When a choice of answers and/or methods is given, mark each attempt. If both methods are valid then M marks can be awarded but any incorrect answer or method would result in marks being lost.

Work not replaced

Erased or crossed out work that is still legible should be marked.

Work replaced

Erased or crossed out work that has been replaced is not awarded marks.

Premature approximation

Rounding off too early can lead to inaccuracy in the final answer. This should be penalised by 1 mark unless instructed otherwise.

Continental notation

Accept a comma used instead of a decimal point (for example, in measurements or currency), provided that it is clear to the examiner that the student intended it to be a decimal point.

Q	Answer	Mark	Comments
1 (a)	colour of flowers	B1	circled or indicated

Q	Answer	Mark	Comments
1 (b)	secondary	B1	circled or indicated

Q	Answer	Mark	Comments
2 (a)	Alternative method 1		
	$153 \div 283 \times 50$ or $153 \times [0.1766, 0.18]$ or $153 \div 5.66$ or $[27.0198, 27.54]$	M1	oe eg 0.54×50
	27	A1	SC1 23
	Alternative method 2		
	$42 \div 283 \times 50$ or 7.4(2...) or 7 and $62 \div 283 \times 50$ or 10.9(5...) or 11 and $49 \div 283 \times 50$ or 8.6(5...) or 9	M1	oe eg using $[0.1766, 0.177]$ for $50 \div 283$
	27	A1	SC1 23
	Additional Guidance		
	SC is for number of adults		
	Use of 0.18 is allowed for method but answer must be 27		
	If they work out number of adults first they must also subtract from 50 for the M1		

Q	Answer	Mark	Comments
2 (b)	$(84 + 85) \div 283 \times 50$ or $(22 + 36 + 62 + 49) \div 283 \times 50$ or $169 \div 283 \times 50$ or $[29.84, 30.42]$ or $(72 + 42) \div 283 \times 50$ or $114 \div 283 \times 50$ or $[20.1, 20.52]$ and subtracts from 50	M1	oe eg works out 84 and 85 eg $(84 + 85) \div 5.66$ eg $169 \times [0.1766, 0.18]$
	30	A1	
	Additional Guidance		
	29 with no working		MOA0

Q	Answer	Mark	Comments
2 (c)	Valid explanation eg It is in proportion/representative of the population. It takes the sample in proportion to the numbers in the groups	B1	oe
	Additional Guidance		
	The sizes for the sample are in the same ratio as the sizes in the groups		B1
	It's suitable because its/the data is split into groups/categories of age and sport		B1
	It's suitable as its split into groups		B0
	It avoids bias		B0
	It's proportional		B0
	It's representative		B0
	Number of people in the sample is fair/equal/even		B0
	It represents the whole sample		B0
	Gives a good proportion of each group		B0
	Because there are different numbers in each group		B0

Q	Answer	Mark	Comments
3 (a)	Point plotted for 2025 from [282 000, 300 000]	B1	must be plotted at correct horizontal distance for 2025
	Additional Guidance		
	Mark point only. This could be the end of the line.		
	The bottom bound is 2 small squares up from 270 000 This is the minimum possible point to be marked		

Q	Answer	Mark	Comments
3 (b)	2015	B1	
	Additional Guidance		
	June 2015		B1

Q	Answer	Mark	Comments
3(c)	Reading at 1999 of [72 000, 78 000]	M1	
	$\frac{281\,585 - \text{their } [72\,000, 78\,000]}{\text{their } [72\,000, 78\,000]} (\times 100)$ or $\frac{281\,585}{\text{their } [72\,000, 78\,000]} - 1$ or [2.61, 2.91] or $\frac{281\,585}{\text{their } [72\,000, 78\,000]} \times 100 (-100)$ or [361, 391.1]	M1	oe their reading must be [60 000, 90 000]
	[261, 291.1] with M2 seen	A1	
	Additional Guidance		
	Rounding of 281 585 eg to 280 000 scores max first M1		
	It is possible to score M0M1A0		

Q	Answer	Mark	Comments
4	Makes an assumption about amount of time spent brushing teeth per day eg 2 minutes twice a day or 4 mins	B1	allow 2 to 10 mins oe eg weekly - allow 14 to 70 mins
	Makes an assumption about the average years a person lives	B1	allow 65 to 90
	their time per day $\times$ days in a year $\times$ their lifespan or their time per week $\times$ weeks in a year $\times$ their lifespan	M1	oe time for their lifespan can be in minutes for M1 allow 360, 364, 365, 365.25 for days in a year allow 48, 50 or 52 if used weeks
	Calculates correct value for their assumptions with answer in hours	A1ft	ft their assumptions must round or truncate to integer number of hours allow further rounding to at least 2sf with correct method seen
	Additional Guidance		
	For method mark allow subtraction of up to 5 years from lifespan for time not brushing their teeth eg as baby or when older Original value should be marked for the B1 for lifespan		
	Extra method should not be ignored for M1 eg also multiplies by 24		M0
	Assumptions must be seen to award the A1 but do not have to be in tolerance		
	Final answer must be in hours		
	Example Assume 2 mins twice a day Assume life span of 80 years $2 \times 2 \times 365 = 1460$ $1460 \times 80 = 116800$ mins $116800 \div 60$ Answer 1950		B1 B1 M1 A1

Q	Answer	Mark	Comments
5	$1 - 0.075$ or 0.925	M1	oe eg percentage
	$\frac{16\,250}{0.925^5}$ or $23\,996.(...)$	M1	accept 24 000 with correct method seen
	7746	A1	SC1 11 004.(...) or 5245 or 5246
	Additional Guidance		
	Rounding or truncating the value of 0.925^5 loses the accuracy mark eg $0.925^5 = 0.68$ $16\,250 \div 0.68 = 23\,897$ $23\,897 - 16\,250 = 7\,647$		M1 M1 A0

Q	Answer	Mark	Comments
6	Mean 52.8(...) or Median [53, 54]	B1	
	SD 18.5(3...) or 19 or 18.4(9...) or 18.5 or IQR [25, 26]	B2	B1 for variance [341.9, 342] or [343, 343.5] or LQ 40 or UQ [65, 66]
	Comparison using mean or median eg the mean/median average for 2023 is higher than 2024 showing that (on average) the 2023 intake performed better.	B1ft	ft their mean/median compared with the same average for 2023
	Comparison using Standard Deviation or IQR eg the SD/IQR for 2024 is lower than for 2023 showing that the 2024 intake results were more consistent	B1ft	ft their SD/IQR compared with the same measure of spread for 2023
	Additional Guidance		
	IQR in correct range but from incorrect LQ or UQ does not score B2 B1 can be awarded if one of LQ and UQ is correct		
	For comparison they must state which intake had the better performance not just 2023 had higher scores eg the mean in 2023 is higher so they did better		B1
	If they only compare using table numbers -Table 1 and Table 2 (not referring to years) then withhold the B1 just the first time it occurs		
	If both averages and/or both spreads are calculated and one value or pair is incorrect then award max B2 for any correct values eg mean and sd correct but median and/or IQR incorrect eg mean correct, median incorrect, sd correct IQR not calculated eg mean and median incorrect, sd correct IQR incorrect		B2 B2 B2
	Comparisons can be eg for mean and then IQR or median and standard deviation		

	Overall the performance was better in 2023 (not clear if its average or consistency and the consistency is poorer)	B0
	Mean is higher in 2023	B0
	2024 marks are less spread out/less varied	B1

Q	Answer	Mark	Comments
7 (a)	Alternative method 1-rearranges formula		
	Makes an assumption about wind speed, v	B1	allow 10 to 18 must be stated not just used in the formula but allow written next to the variables given in the question
	Makes an assumption about the capacity factor, C	B1	allow 25% to 40% or 0.25 to 0.4(0) must be stated not just used in the formula but allow written next to the variables given in the question
	Works out power output in watts or MW per wind turbine eg $\frac{0.75 \times 1200}{80} (\times 1\,000\,000)$ or $(750\,000 \times 1200) \div 80$ or 11 250 000 (watts) or 11.25 (MW)	M2	oe multiplication by 0.75, division by 80 may be done after subbing in formula incorrect conversion to watts can gain M1 for one valid step M1 one valid step $1200 \div 80$ or 15 (watts per turbine per home) or $750\,000 \div 80$ or 9375 or $750\,000 \times 1200$ or 900 000 000 (total watts) or 0.75×1200 or 900 (total MW)

7 (a) cont'd	Subs in power output formula for their wind speed, capacity factor and power per turbine in watts eg using wind speed of 12 and capacity factor 0.35 their $11\,250\,000 = \frac{1}{2} \times 1.23 \times A \times 12^3 \times 0.35$ or their $11.25 \times 1\,000\,000 = \frac{1}{2} \times 1.23 \times A \times 12^3 \times 0.35$	M2	oe their 1 125 000 can be their 1200 000 000 or their 900 000 000 but must be in watts unless conversion is seen later using 750 000 (homes instead of power output) is counted as one error allow πr^2 used for A but blade length used is one error M1 subs in formula with one error or omission eg $11\,250\,000 = 1.23 \times A \times 12^3 \times 0.35$ eg $11.25 = \frac{1}{2} \times 1.23 \times A \times 12^3 \times 0.35$
	Rearranges the formula for A with or without values subbed in eg for values in previous mark $A = 11\,250\,000 \div 371.952$ or 30 246	M1	ft their v, C and power output allow rounding to at least 2 sf allow watts or MW may imply previous M2 M2 for their assumptions
	Uses area of a circle formula with their 30 246 eg $\pi r^2 = 30\,246$	M1	implied by correct rearrangement for r
	rearranges for r eg $r = \sqrt{\frac{30\,246}{\pi}}$	M1	ignore doubling to diameter as fw
	Correct blade length(radius) for their assumptions eg 98	A1ft	ft their assumptions with all method marks scored must have used correct rearranged formula for r must be rounded at least to an integer

7(a) cont'd	Alternative method 2 – Trial and improvement		
	Makes an assumption about wind speed, v	B1	allow 10 to 18 must be stated not just used in the formula but allow written next to the variables given in the question
	Makes an assumption about the capacity factor, C	B1	allow 25% to 40% or 0.25 to 0.4(0) must be stated not just used in the formula but allow written next to the variables given in the question
	Works out power output in watts or MW per wind turbine eg $\frac{0.75 \times 1200}{80} (\times 1\,000\,000)$ or $(750\,000 \times 1200) \div 80$ or 11 250 000 (watts) or 11.25 (MW)	M2	oe multiplication by 0.75, division by 80 may be done after subbing in formula M1 one valid step $1200 \div 80$ or 15 (watts per turbine per home) or $750\,000 \div 80$ or 9375 or $750\,000 \times 1200$ or 900 000 000 (total watts) or 0.75×1200 or 900 (total MW)
	Uses area of a circle formula for any blade length eg $\pi \times 90^2$ or 25 446 or 25 447	M1	must show their blade length
	Subs in power output formula for their wind speed, capacity factor and any area/blade length eg using wind speed of 12 and capacity factor 0.35 and blade 90(m) $\frac{1}{2} \times 1.23 \times \pi \times 90^2 \times 12^3 \times 0.35$	M2	implies previous M1 their area cannot be their power output M1 subs in formula with one error or omission eg $1.23 \times \pi \times 90^2 \times 12^3 \times 0.35$ eg $\frac{1}{2} \times 1.23 \times 90^2 \times 12^3 \times 0.35$
	Uses a blade length that gives a power output of [11 200 000, 11 400 000] watts or [11.2, 11.4] MW	M2dep	dep on all previous method marks using their assumptions for wind speed and capacity factor M1 uses a blade length that gives a power output of [11 000 000, 12 000 000] watts or [11, 12] MW

	Correct blade length for their assumptions eg 98	A1ft	ft their assumptions with all method marks scored must be rounded at least to an integer
	Additional Guidance		
	Some steps may be done in a different order eg considering the 80 turbines may be included when subbing into the formula eg for wind speed 12 and capacity factor 0.35 $900\,000\,000 = \frac{1}{2} \times 1.23 \times A \times 12^3 \times 0.35 \times 80$ This is an equivalent method and recovers the division by 80 so gains M2M2 Division by 80 or equivalent must be done before rearranging their area for r		
	Assumptions must be stated so just writing wind speeds and capacity factor in the formula does not gain the B marks However allow the values to be clearly shown next to wind speed and capacity factor next to $v =$ and $C =$ in the question or stated as $v =$ and $C =$ in the working lines		
	Correct total for intermediate steps must come from assumptions seen eg assumes wind speed = 12 and CF = 0.35 Then $11\,250\,000 = 371.952 \times A$ is correct for their assumptions		
	Alt 2 eg Assuming wind speed is 8 Assuming capacity factor is 25% Power output should be 11 250 000 Assume blade length is 100 $\frac{1}{2} \times 1.23 \times \pi \times 100^2 \times 8^3 \times 0.25 = 2\,473\,061$ too low Try blade length 200 = 9 892 246 still too low Try 215, power 11 431 728 that's enough Answer 215 (one trial is within wider range for M1, actual blade length is 213.28)	B0 B1 M2 M1M2 M1dep A0	
	Alt 2 A single T&I method that leads to the correct power output scores all method marks eg Assume wind speed is 14 and capacity factor 40% 11 250 000 (watts) needed	B2 M2	

	$\frac{1}{2} \times 1.23 \times \pi \times 73^2 \times 14^3 \times 0.4 (= 11\,300\,946.19)$ Answer 73 (Actual blade length is 72.835...)	M1M2M2 A1ft
	Working in MW can lead to a correct answer if either they divide the RHS of the formula by 1 000 000 or they divide their radius by $\sqrt{1\,000\,000}$ They can retrospectively gain the 2 nd M1 of the M2 in Alt 11 for subbing in formula	
	A mix of Alts may be possible eg trial of areas (Alt 2) then rearranges their area for r (Alt 1)	

Q	Answer	Mark	Comments
7 (b)	Acceptable explanation for how blade length is affected Examples If my assumption for the wind speed was too low then my estimate for the blade length would be too high or If the capacity factor was higher than my assumption then the blade length would be shorter than my answer	B1	must relate to their assumptions
	Additional Guidance		
	The wind speeds may vary which would change my answer		B0
	If the capacity factor was more/less than I assumed then the blade length would have to be smaller/larger		B1
	If the wind speeds were lower the turbines would not move as fast so less electricity would be generated		B0

Q	Answer	Mark	Comments
8 (a)	Alternative method 1		
	$(114\,900 - 100\,000) \div 2 = 7450$ and $12\,570 - 7450 = 5120$	B2	B1 $(114\,900 - 100\,000) \div 2$ accept 14 900 for $(114\,900 - 100\,000)$
	Alternative method 2		
	$[12\,570 \times 2 - (114\,900 - 100\,000)] \div 2$ $= 5120$	B2	B1 $[12\,570 \times 2 - (114\,900 - 100\,000)]$ accept 14 900 for $(114\,900 - 100\,000)$
	Additional Guidance		
	$12\,570 - 5120 = 7450$		B0

Q	Answer	Mark	Comments
8 (b)	Alternative method 1		
	114 900 – 5120 – 37 700 or 109 780 – 37 700 or 72 080	M1	oe calculating amount for higher rate tax
	their 72 080 $\times$ 0.4 or 28 832 and 37 700 $\times$ 0.2 or 7540	M1	oe their 72 080 < 114 900 and not 37 700 total tax of 36 372 is M2
	114 900 – 50 270 $\times$ 0.02 or 64 630 $\times$ 0.02 or 1292.6(0)	M1	oe higher rate NI
	50 270 – 12 570 $\times$ 0.08 or 37 700 $\times$ 0.08 or 3016	M1	oe standard NI total NI of 4308.6(0) is 3rd and 4th M1
	their 28 832 + their 7540 + their 1292.6(0) + their 3016	M1	oe must be higher and standard tax and NI may be seen embedded in subtraction
	40 680.6(0)	A1	total deductions implied by net salary of 74 219.4(0)
	114 900 – their 40 680.6(0) or 74 219.4(0)	M1	works out net salary their 40 680.6(0) < 114 900
	their 74 219.4(0) $\div$ 114 900 or [0.64, 0.65] or their 74 219.4(0) $\div$ 2 $\times$ 3 or $\frac{2}{3} \times 114 900$ or 76 600	M1	oe percentage their 74 219.4(0) must be their net salary may be seen earlier in working
	[0.64, 0.65] and Yes or 111 329.(1) and Yes or their 74 219.(40) and 76 600 and Yes	A1ft	oe percentage ft their 40 680.6(0) for net salary

Q	Answer	Mark	Comments
8 (b) cont'd	Alternative method 2		
	114 900 – 5120 – 37 700 or 109 780 – 37 700 or 72 080	M1	oe calculating amount for higher rate tax
	their 72 080 $\times$ 0.4 or 28 832 and 37 700 $\times$ 0.2 or 7540	M1	oe their 72 080 < 114 900 and not 37 700 total tax of 36 372 is M2
	114 900 – 50 270 $\times$ 0.02 or 64 630 $\times$ 0.02 or 1292.6(0)	M1	oe higher rate NI
	50 270 – 12 570 $\times$ 0.08 or 37 700 $\times$ 0.08 or 3016	M1	oe standard NI total NI of 4308.6(0) is 3rd and 4th M1
	their 28 832 + their 7540 + their 1292.6(0) + their 3016	M1	oe must be higher and standard tax and NI may be seen embedded as subtraction from 114 900
	40 680.6(0)	A1	total deductions implied by net salary of 74 219.4(0)
	Full method to enable comparison eg their 40 680.6(0) $\div$ 114 900 or 0.35(4..) and either 1 – their 0.35(4...) or [0.64, 0.65] or $1 - \frac{2}{3}$ or $\frac{1}{3}$ or 0.33 or $1 - \frac{2}{3}$ or $\frac{1}{3}$ or 0.33 and 114 900 $\div$ 3 or 38 300	M2	oe valid comparison their 40 680.6(0) < 114 900 and must be their total deductions M1 one step towards comparison their 40 680.6(0) $\div$ 114 900 or 0.35(4..) or $1 - \frac{2}{3}$ or $\frac{1}{3}$ or 0.33 allow equivalent percentages

8(b) cont'd	[0.64, 0.65] and Yes or 0.35(4..) and 0.33 and Yes or their 40 680.6(0) and 38 300 and Yes	A1ft	ft their 40 680.6(0)
	Additional Guidance		
	Yes can be implied eg She is correct or Her net salary is less than $\frac{2}{3}$ of her gross salary However do not accept just inequality signs as implying Yes eg $74\,219.(40) < 76\,600$		
	Ignoring the reduced personal allowance and using 12 570 can gain up to 7 marks The first method mark and the first A1 are not scored eg $114\,900 - 12\,570 - 37\,700 = 64\,630$ $64\,630 \times 0.4 = 25\,852$ $37\,700 \times 0.2 = 7540$ (total tax = 33 392) All other marks except the first A1 are then possible		M0 M1
	Using the starting value for NI as $114\,900 - 5120$ or 109780 can gain up to 7 marks eg Higher tax 28 832 and standard tax 7540 Higher NI $(109\,780 - 50\,270) \times 0.02 = 1190.2(0)$ Standard NI = 3016 Total deductions $28\,832 + 7540 + 1190.2(0) + 3016 = 40\,578.2(0)$ $114\,900 - 40\,578.2(0) = 74\,321.8(0)$ $\frac{2}{3} \times 114\,900 = 76\,600$ She is correct		M1M1 M0 M1 M1A0 M1 M1A1ft

Q	Answer	Mark	Comments
9 (a)	Alternative method 1 – cumulative frequency diagram		
	9, 37, 59, 77, 80	B1	correct cumulative frequencies implied by correct heights
	Linear scales for both axes and correct labelling of axes	B1	cf scale starting from zero up to at least 80 but condone zero not labelled high jump height scale may start from 110 vertical label cumulative frequency or cf horizontal scale must have cm but can be height/ height jumped/h
	Correct cumulative frequency graph joined with lines or smooth curve	B2ft	$\pm \frac{1}{2}$ square ft their cf values which must be increasing ignore any line/curve to the left of their point at cf 9 B1ft all their cf heights correct and joined with lines/smooth curve but plotted at incorrect horizontal position or B1ft plotted at upper class values and joined with lines/smooth curve with at least 3 of their cf heights correct or B1ft all their points correct but no lines/curve or poor lines/curve
	Alternative method 2 – histogram		
	At least 4 correct frequency densities 0.9, 1.4, 2.2, 1.8, 0.1	B1	may be implied by bar heights
	Linear scales for both axes and correct labelling of axes	B1	fd scale starting from zero up to at least their maximum fd but condone zero not labelled high jump height scale may start from 110 vertical label frequency density or fd horizontal scale must have cm but can be height/ height jumped/h
	All bars correct	B2	B1 3 or 4 bars correct

Additional Guidance is on the next page

9(a) cont'd	Additional Guidance	
	Scales and labelling mark can only be given for a cf scale or an fd scale -not for number of students/frequency scales	
	For cf graph ignore any curve/line below a high jump height of 110 Accept a horizontal line drawn from their final point, but do not accept a continuation of the curve or polygon.	
	A poor cf curve is one that misses any points by more than $\frac{1}{2}$ square or is a double/feathered line or goes above 80	
	For the histogram the lines do not need to be ruled. Use judgement to mark intention	
	For both graphs ignore extra horizontal/vertical lines which may be their working for part (b)	
	If a cf histogram is drawn with a cf curve on top mark the curve Labels should be for cf	

Q	Answer	Mark	Comments
9 (b)	High jump From a cf graph for high jump Reads their correct number of students above 145 cm or from table $\frac{1}{2} \times 22 + 18 + 3$ or 32	M1	ft increasing graph
	Long jump Any bar frequency calculated 10×0.9 or 9 or 10×1.8 or 18 or 15×2.4 or 36 or 20×2.4 or 48 or 5×2.4 or 12 or 20×1.5 or 30 or 30×0.5 or 15	M1	oe eg counting squares may be on diagram can be the quarter or three-quarter of 410 to 430 bar
	$5 \times 2.4 + 20 \times 1.5 + 30 \times 0.5$ or $12 + 30 + 15$ or $120 - (10 \times 0.9 + 10 \times 1.8 + 15 \times 2.4)$ or 57	M1	implies previous M1 oe eg counting squares if working with values under 425 the percentage may be worked out first then subtracted (so method marks in reverse order)
	(High jump =) their $32 \div 80 (\times 100)$ or (Long jump =) their $57 \div 120 (\times 100)$	M1	oe their 32 is their reading from the cf graph or from the table ft their high jump and long jump values

	From histogram 40(%) and 47.(5)(%) and long jump or from cf graph [37.5, 45](%) and 47.(5)(%) and long jump	A2	oe eg decimals A1 from histogram 40(%) and 47.(5)(%) or from cf graph [37.5, 45](%) and 47.(5)(%) or A1ft one value correct with correct ft decision for their two values
	Additional Guidance		
	The bounds for the cf graph final percentages are for using readings of [30, 36] For any other values only the M1 for their $32 \div 80 (\times 100)$ is possible		

Q	Answer	Mark	Comments
10	Alternative method 1		
	$\frac{20.8(0)}{5200} (\times 100)$ or 0.4(%) or 0.004	M1	oe implied by 4.8 or 0.048
	$[(1 + \frac{0.048}{12})^{12} - 1] (\times 100)$ or $[(1 + 0.004)^{12} - 1] (\times 100)$ or 0.049(07)	M2dep	oe M1 one error in formula or their 0.004×12 or 0.048 oe
	4.9(07...) or 4.91	A1	
	Alternative method 2		
	$5220.8(0) \div 5200$ or 1.004	M1	oe
	$(1.004^{12} - 1) (\times 100)$ or 0.049(07)	M2	oe M1 1.004^{12} or 1.04907
	4.9(07...) or 4.91	A1	
	Additional Guidance		
	12 not correct in both places counts as one error if both new values are the same		
	Use the Alt that favours the student		